Sensors

Preface

- We will take a closer look at different kind of sensors used in the remote sensing process.

- characteristics of remote sensing platforms and sensors, the data they collect.

- touch briefly on how those data are processed once they have been recorded by the sensor.

Platform of sensors



In order for a sensor to collect and record energy reflected or emitted from a target or surface, it must reside on a stable **platform** above the target or surface being observed.

Platforms for remote sensors may be situated on the ground, on an aircraft or balloon (or some other platform within the Earth's atmosphere), or on a spacecraft or satellite outside of the Earth's atmosphere.

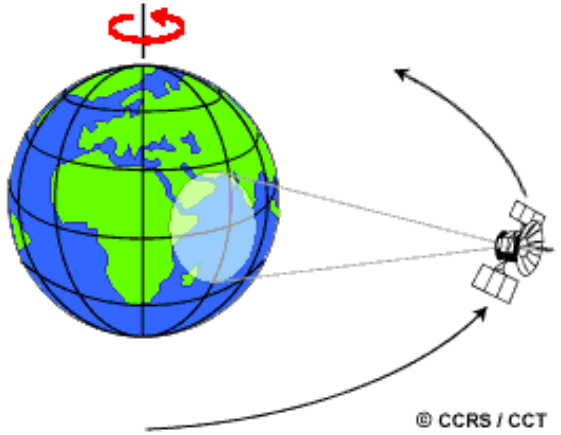
- **Ground-based sensors** are often used to record detailed information about the surface which is compared with information collected from aircraft or satellite sensors.

- In some cases, this can be used to better characterize the target which is being imaged by these other sensors, making it possible to better understand the information in the imagery.

- **Aerial sensors** are mounted primarily on stable wing **aircraft**, although helicopters are occasionally used.

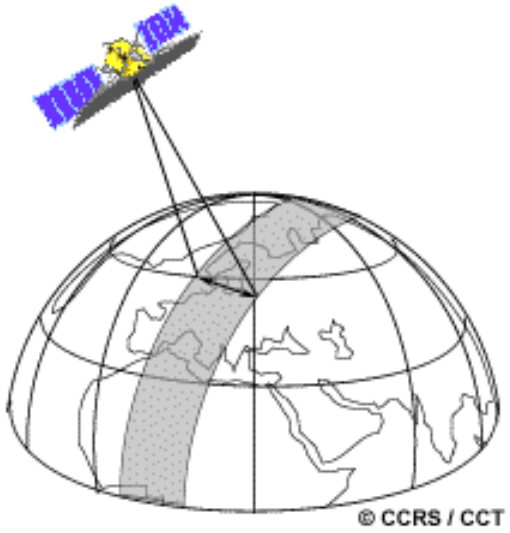
- UAVs & Satellites are often used to collect very detailed images and facilitate the collection of data over virtually any portion of the Earth's surface at any time.

Satellite Characteristics: Orbits and Swaths



The path followed by a satellite is referred to as its **orbit**. Satellite orbits are matched to the capability and objective of the sensor(s) they carry.

Orbit selection can vary in terms of altitude (their height above the Earth's surface) and their orientation and rotation relative to the Earth.



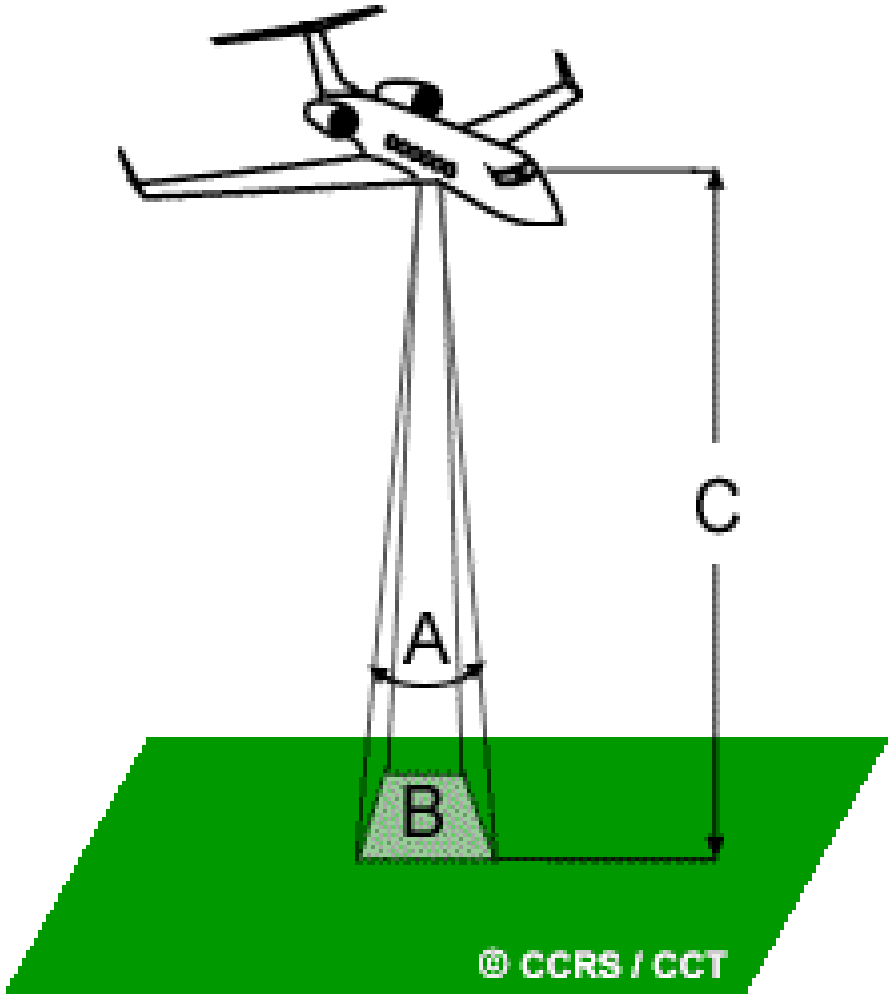
As a satellite revolves around the Earth, the sensor "sees" a certain portion of the Earth's surface. The area imaged on the surface, is referred to as the **swath**.

Imaging swaths for space borne sensors generally vary between tens and hundreds of kilometres wide.

Spatial Resolution, Pixel Size, and Scale

- For remote sensing instruments, the distance between the target being imaged and the platform, plays a large role in determining the detail of information obtained and the total area imaged by the sensor.

- Sensors onboard platforms far away from their targets, typically view a larger area, but cannot provide great detail.



- The detail discernible in an image is dependent on the **spatial resolution** of the sensor and refers to the size of the smallest possible feature that can be detected.

- Spatial resolution of passive sensors depends primarily on their **Instantaneous Field of View (IFOV)**.

- The IFOV is the angular cone of visibility of the sensor (A) and determines the area on the Earth's surface which is "seen" from a given altitude at one particular moment in time (B).

- The size of the area viewed is determined by multiplying the IFOV by the distance from the ground to the sensor (C). This area on the ground is called the **resolution cell** and determines a sensor's maximum spatial resolution.

- Most remote sensing images are composed of a matrix of picture elements, or **pixels**, which are the smallest units of an image.

- Image pixels are normally square and represent a certain area on an image.

- Pixel size and spatial resolution are different. They are not interchangeable.

- If a sensor has a spatial resolution of 20 metres and an image from that sensor is displayed at full resolution, each pixel represents an area of 20m x 20m on the ground.
  - In this case the pixel size and resolution are the same.

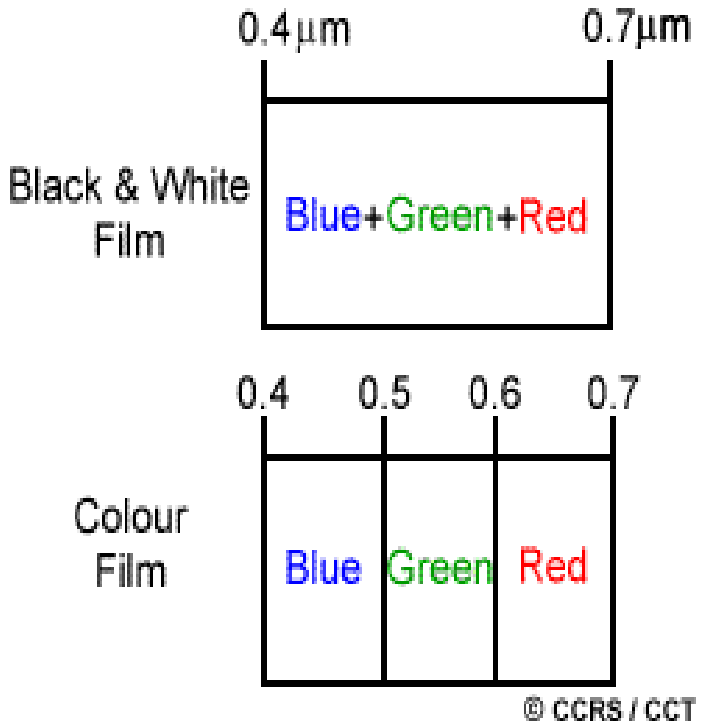
- However, it is possible to display an image with a pixel size different than the resolution.

- Images where only large features are visible are said to have **coarse or low resolution.** In **fine or high resolution** images, small objects can be detected.

Spectral Resolution

Spectral resolution describes the ability of a sensor to define fine wavelength intervals.

The finer the spectral resolution, the narrower the wavelength range for a particular channel or band and vice-versa.



Black and white film records wavelengths extending over much, or all of the visible portion of the electromagnetic spectrum. Its **spectral resolution** is fairly coarse.

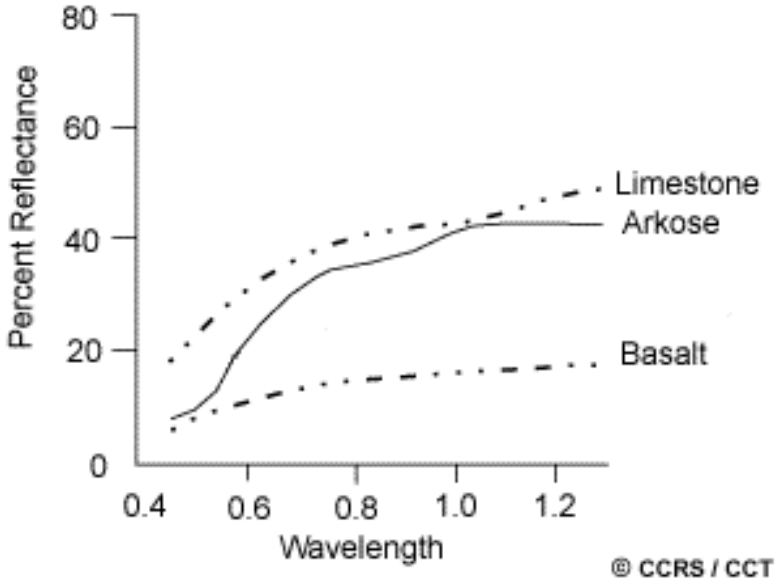
Colour film is also sensitive to the reflected energy over the visible portion of the spectrum, but has higher spectral resolution.

- **Spectral response** and **spectral emissivity curves** which characterize the reflectance and/or emittance of a feature or target over a variety of wavelengths.

- Different classes of features and details in an image can often be distinguished by comparing their responses over distinct wavelength ranges.

- Broad classes, such as water and vegetation, can usually be separated using very broad wavelength ranges - the visible and near infrared.

- Other more specific classes, such as **different rock types**, may not be easily distinguishable using either of these broad wavelength ranges and would require comparison at much finer wavelength ranges to separate them. Thus, we would require a sensor with higher **spectral resolution**.



- Many remote sensing systems record energy over several separate wavelength ranges at various spectral resolutions. These are referred to as **multi-spectral sensors**.

- Advanced multi-spectral sensors called **hyperspectral** sensors, detect hundreds of very narrow spectral bands throughout the visible, near-infrared, and mid-infrared portions of the electromagnetic spectrum. Their very high spectral resolution facilitates fine discrimination between different targets based on their spectral response in each of the narrow bands.

Radiometric Resolution

- While the arrangement of pixels describes the spatial structure of an image, the radiometric characteristics describe the actual information content in an image.

- Every time an image is acquired on film or by a sensor, its sensitivity to the magnitude of the electromagnetic energy determines the **radiometric resolution**.

- The radiometric resolution of an imaging system describes its ability to discriminate very slight differences in energy The finer the radiometric resolution of a sensor, the more sensitive it is to detecting small differences in reflected or emitted energy.





Radiometric Resolution

Radiometric resolution: Sensitivity of the sensor to the magnitude of the electromagnetic energy

- How many grey levels are measured between pure black (no reflectance) to pure white (maximum reflectance)

- The finer the radiometric resolution of a sensor the more sensitive it is in detecting small differences in the energy

- The finer the radiometric resolution of a sensor the system can measure more number of grey levels

- Radiometric resolution is measured in Bits
  - ➢ Each bit records an exponent of power 2
- Maximum number of brightness levels available depends on the number of bits used in representing the recorded energy

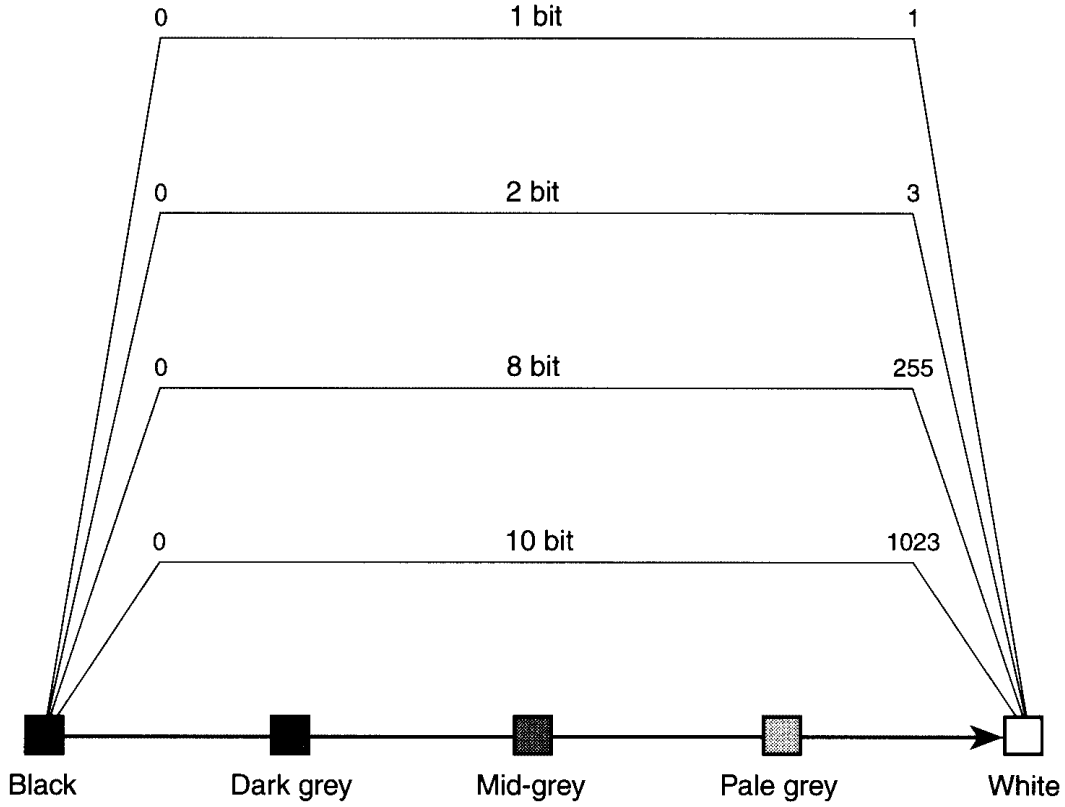
Radiometric resolution and the corresponding brightness levels available

	Radiometric resolution	Number of levels	Example
Poor resolution	1 bit	2^1 – 2 levels	
	7 bit	2^7 – 128 levels	IRS 1A & 1B
	8 bit	2^8 – 256 levels	Landsat TM
High resolution	11 bit	2^{11} – 2048 levels	NOAA-AVHRR

- Tones in an image vary from black to white
- Black → Digital Number = 0 → No reflectance
- White → Digital Number is the maximum

  =1 for a 1-bit data

  =255 for a 8-bit data



- Finer radiometric resolution
  - More the number of grey levels
  - More details can be captured in the image

- Finer radiometric resolution
  - Increases the data storage requirements





- Digital number (DN) depends on the number of brightness levels
  - Lower DN value in a coarse resolution image
  - Higher DN value in a fine resolution image

Example

- DNs recorded by the 3-bit system range from 0 to 7
- This range is equivalent to 0-63 for the 6 bit system

0	1	2	3	4	5	6	7	(3 bit)
0	9	18	27	36	45	54	63	(6 bit)

DN of 45 (6-bit) → 5 (3-bit)

To compare two images, their radiometric resolution should be the same

- Imagery data are represented by positive digital numbers which vary from 0 to (one less than) a selected power of 2. This range corresponds to the number of bits used for coding numbers in binary format.

- Each bit records an exponent of power 2 (e.g. 1 bit=2^1=2). The maximum number of brightness levels available depends on the number of bits used in representing the energy recorded.

- Thus, if a sensor used 8 bits to record the data, there would be 2^8=256 digital values available, ranging from 0 to 255. However, if only 4 bits were used, then only 2^4=16 values ranging from 0 to 15 would be available. Thus, the radiometric resolution would be much **less**.

- Image data are generally displayed in a range of grey tones, with black representing a digital number of 0 and white representing the maximum value (for example, 255 in 8-bit data).

- By **comparing a 2-bit image with an 8-bit image**, we can see that there is a large difference in the level of detail discernible depending on their radiometric resolutions.

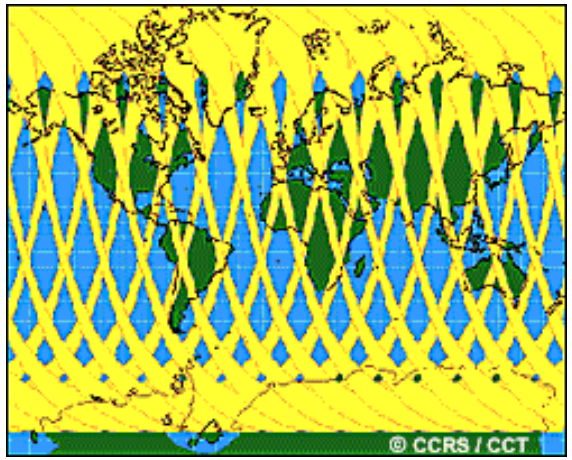
Temporal Resolution

- Number of times an object is sampled or How often data are obtained for the same area
  - ❖ The absolute temporal resolution of a remote sensing system to image the same area at the same viewing angle a second time is equal to the repeat cycle of a satellite.
  - ❖ The repeat cycle of a near polar orbiting satellite is usually several days

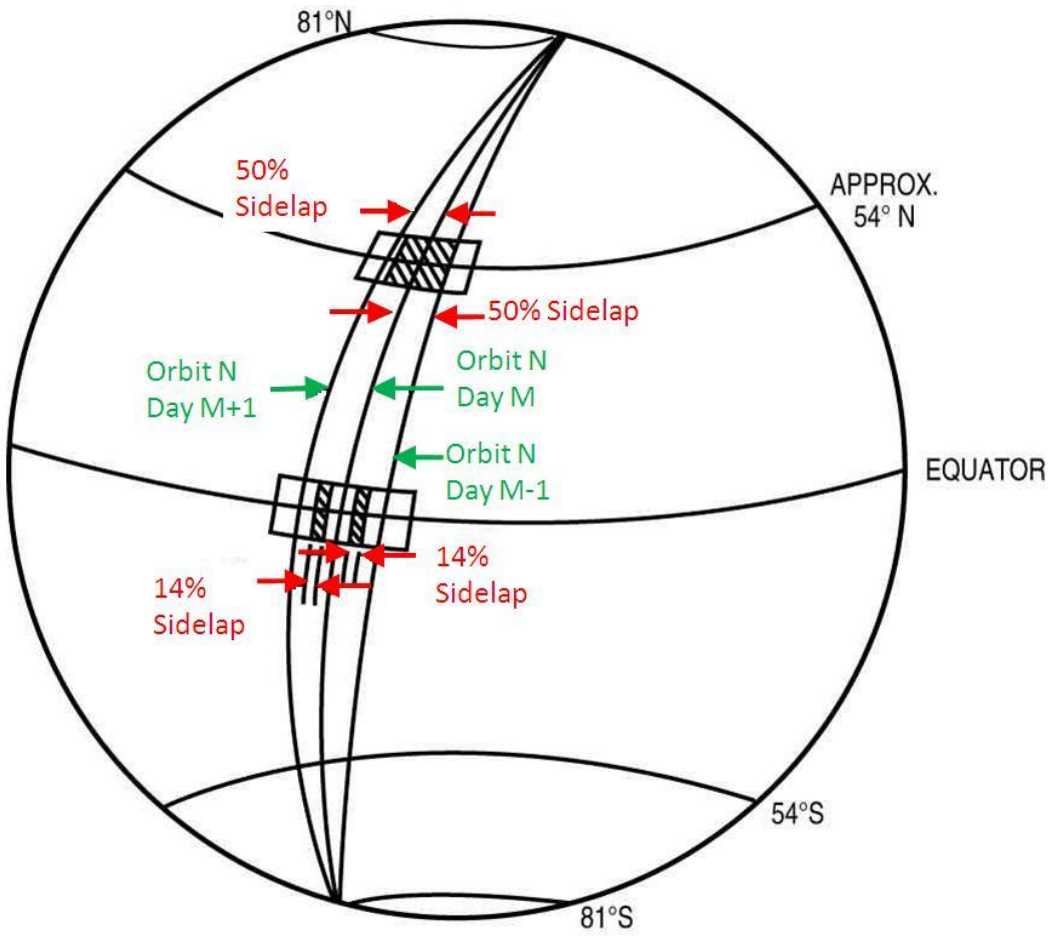
    Example: 24 days for IRS-1C and Resourcesat-2, 18 days for Landsat, 14days for IKONOS
  - ❖ Actual temporal resolution (or revisit period) of a sensor depends on
    - ➢ The satellite/sensor capabilities
    - ➢ Swath overlap and Latitude

Swath Overlap and Latitude

- Sidelap in the swaths of the adjacent orbits increases the frequency of imaging
- Sidelap increases with latitude, increasing the frequency of images available for the polar region



Paths of a Typical Near-Polar Satellite



Towards the polar region, satellite orbits come closer to each other. More frequent images are available for the polar region

Satellite Capabilities and Temporal Resolution

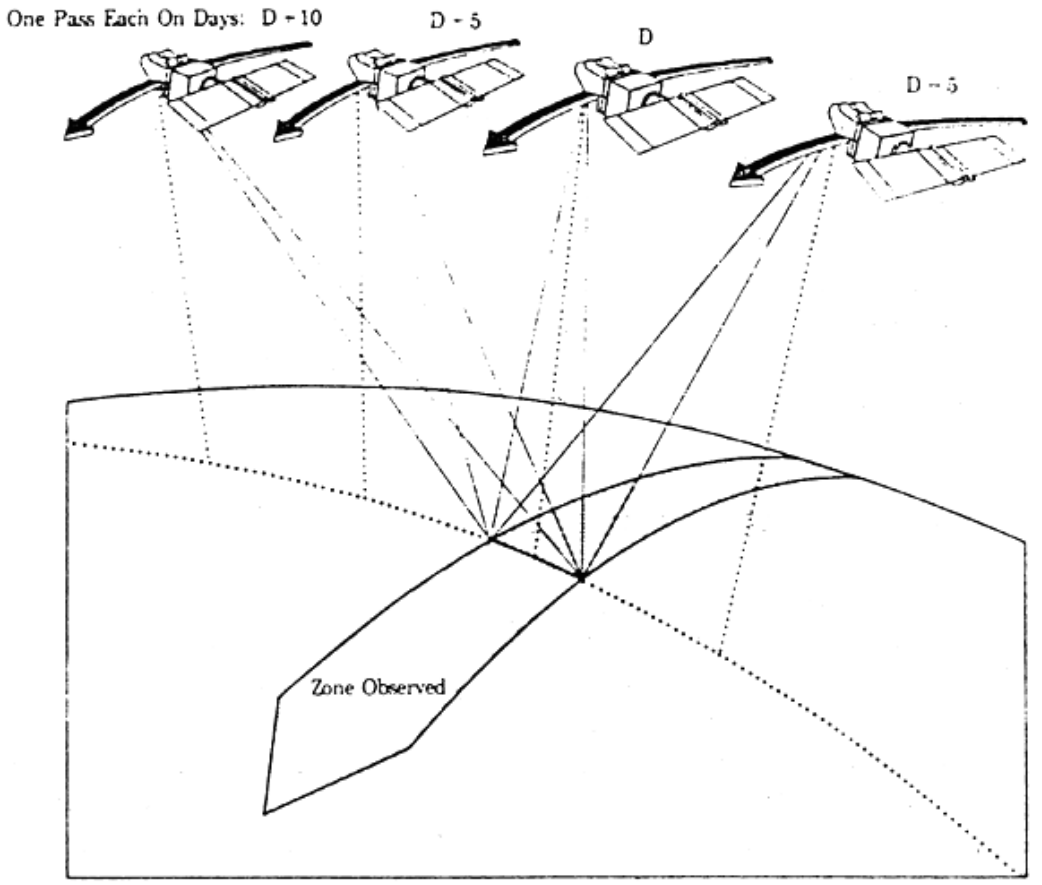
- More frequent imaging is possible by off-nadir viewing capabilities

Example : IKONOS

Sensor characteristics: Pointable optics

Repeat cycle : 14 days

Revisit period : 1-3 days



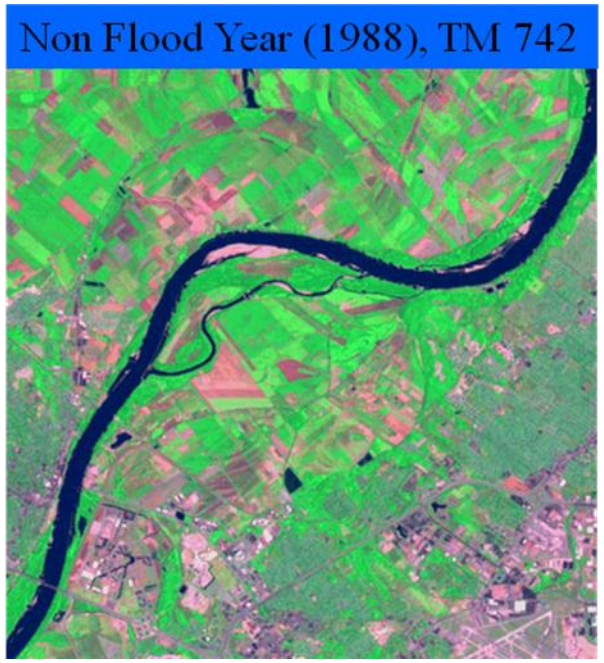
Importance of Temporal Resolution

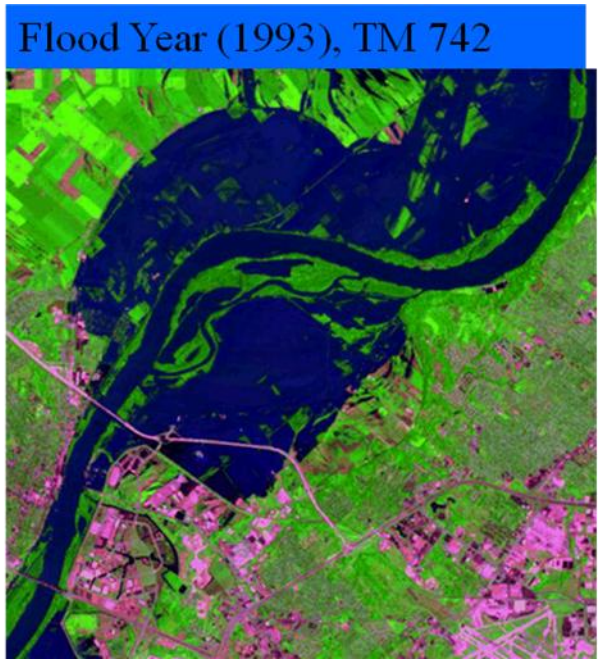
- Images at different time periods show the variation in the spectral characteristics of different features over time
- Applications
  - Land use/ land cove classification
  - Temporal variation in land use / land cover
  - Monitoring of a dynamic events like
    - Cyclone
    - Flood
    - Volcano
    - Earthquake

Flood Studies

- Satellite images before and after the flood event help to identify the aerial extent of the flood during the progress and recession of a flood event

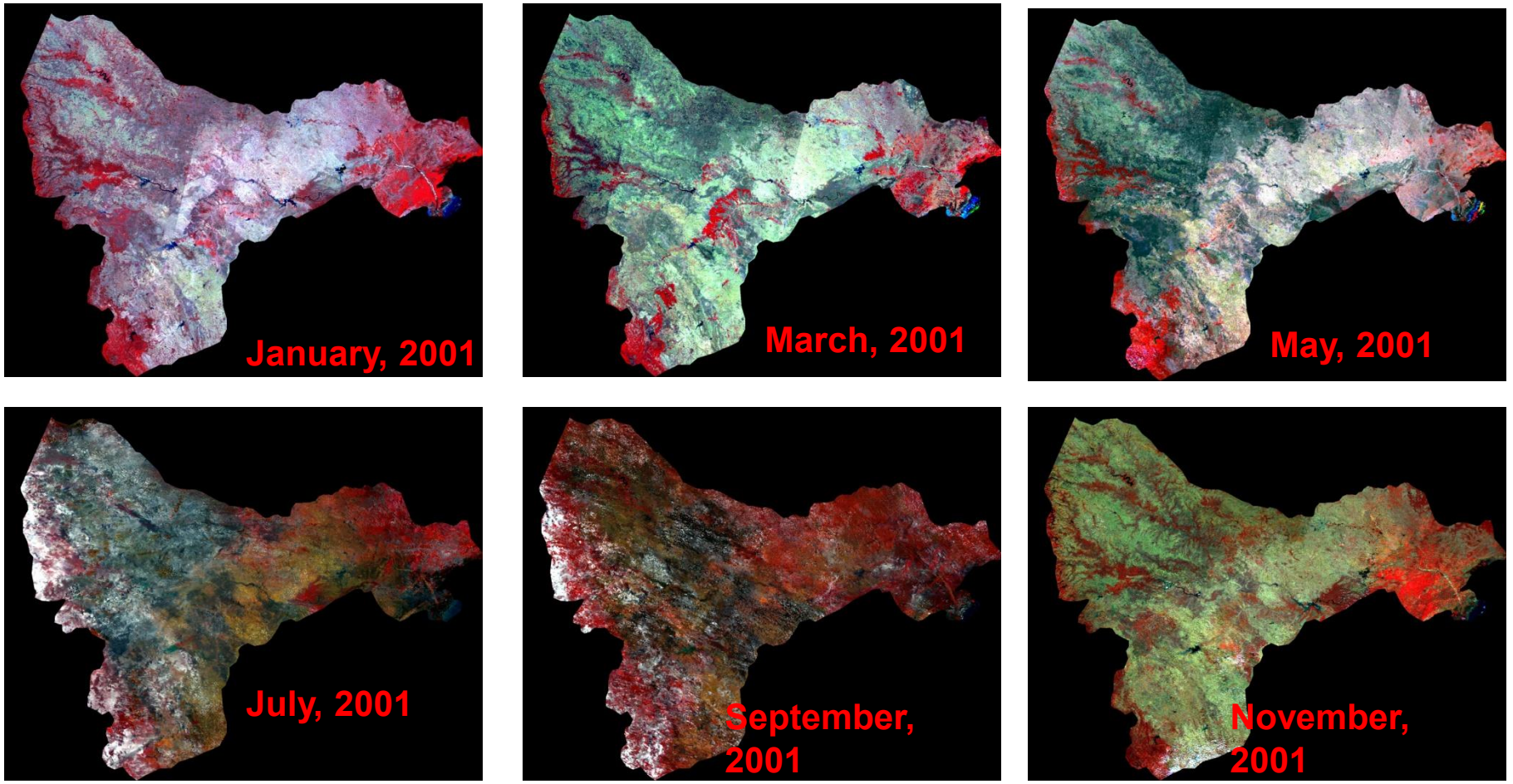
Landsat TM images of the Mississippi River taken during a normal period and during the great flood of 1993





Land Use/ Land Cover Classification:

MODIS data product for the Krishna River Basin



FCC (RGB): 2,1,6 (NIR, red, MIR1) Krishna river basin, India

Signal-to-Noise Ratio

- Signal-to-noise ratio (SNR) depends on strength of signal and the noise of the system
  - Signal (say reflectance)
  - Noise from aberrations in the electronics, moving parts or defects in the scanning system
- Higher the SNR → Differentiation of the noise from the actual signals is easier

- Finer spatial, spectral and radiometric resolutions of a system may decrease the SNR to such an extent that the data may not be reliable
  - Higher spectral and spatial resolution reduces the energy (signal strength) → reduces the SNR
  - Finer radiometric resolution →larger number of grey levels
    - If the difference in the energy level between the two levels is less than the noise, reliability of the recorded grey level diminishes.

Trade-off Between Resolutions

Fine spatial resolution → small IFOV → less energy

- Difficult to detect fine energy differences → Poor radiometric resolution
- Poor spectral resolution

Narrow spectral bands →High spectral resolution → Less energy

- Difficult to detect fine energy differences → Poor radiometric resolution
- Poor spatial resolution

Wide spectral band → Poor spectral resolution→ more reflected energy

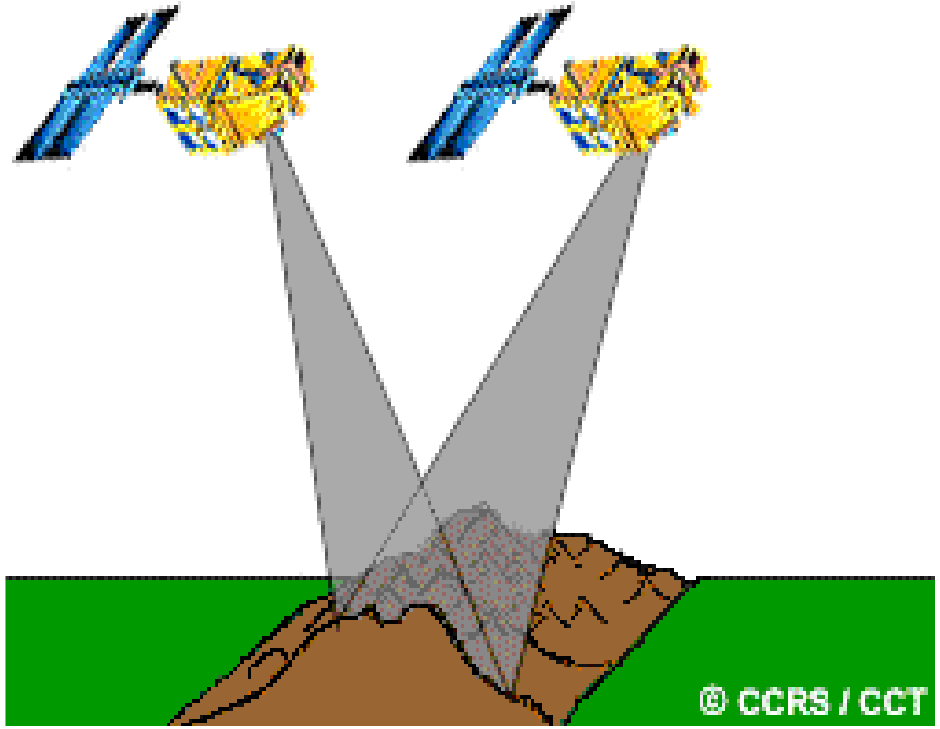
- Good spatial resolution
- Good radiometric resolution

These three types of resolutions must be balanced against the desired capabilities and objectives of the sensor.

Land Use/ Land Cover Classification

- Temporal variation in the spectral signature can be estimated
- Presence of features over time can be identified
- Continuous change in the vegetation characteristics can be monitored
    - Used to classify the crop types viz., perennial crops, long or short duration crops

Temporal Resolution



- **Temporal resolution** of a remote sensing system is the period to image the exact same area at the same viewing angle at two successive time.
- However, because of some degree of overlap in the imaging swaths of adjacent orbits for most satellites and the increase in this overlap with increasing latitude, some areas of the Earth tend to be re-imaged more frequently.
- Also, some satellite systems are able to **point their sensors to image the same area** between different satellite passes separated by periods from one to five days.

- The ability to collect imagery of the same area of the Earth's surface at different periods of time is one of the most important elements for applying remote sensing data.

- Spectral characteristics of features may change over time and these changes can be detected by collecting and comparing **multi-temporal** imagery.

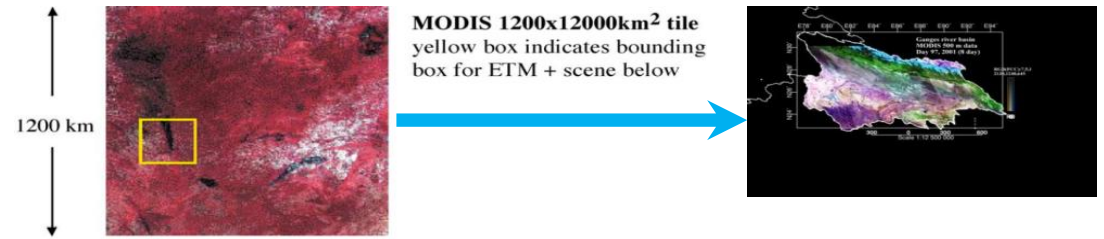
- The time factor in imaging is important when:

  - persistent clouds offer limited clear views of the Earth's surface (often in the tropics)
  - short-lived phenomena (floods, oil slicks, etc.) need to be imaged
  - multi-temporal comparisons are required (e.g. the spread of a forest disease from one year to the next)
  - the changing appearance of a feature over time can be used to distinguish it from near similar features (wheat / maize)

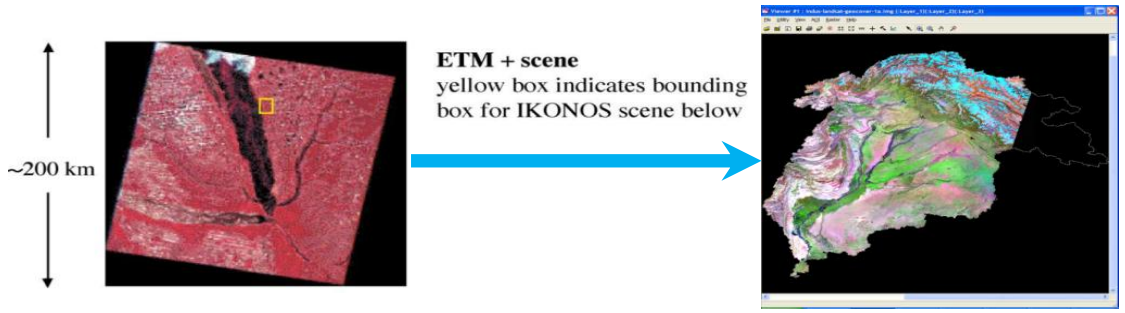
Classes of Spatial Resolution

- Low resolution systems
  - Spatial resolution > 1km
  - MODIS, AVHRR
- Medium resolution systems
  - Spatial resolution is 100m – 1km
  - IRS WiFS (188m), Landsat TM–Band 6 (120m), MODIS–Bands 1-7 (250-500m)
- High resolution systems
  - Spatial resolution approximately in the range 5-100m
  - Landsat ETM+ (30m), IRS LISS-III (23m MSS, 6m Panchromatic), IRS AWiFS (56-70m), SPOT 5(2.5-5m Panchromatic)
- Very high resolution systems
  - Spatial resolution less than 5m
  - GeoEye (0.45m for Panchromatic, 1.65m for MSS), IKONOS (0.8-1m Panchromatic), Quickbird (2.4-2.8 m)

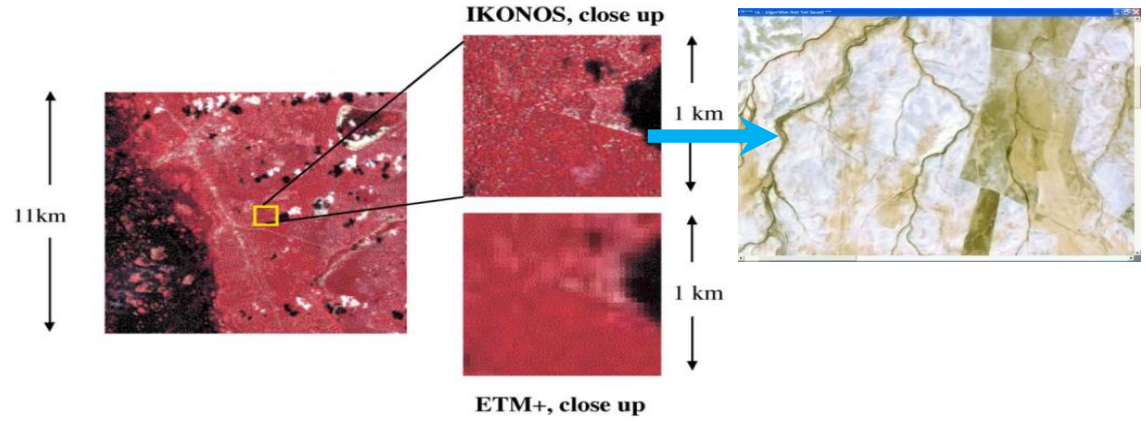
Spatial Resolutions and Scale of Applicability



Ganges basin from MODIS 500 m

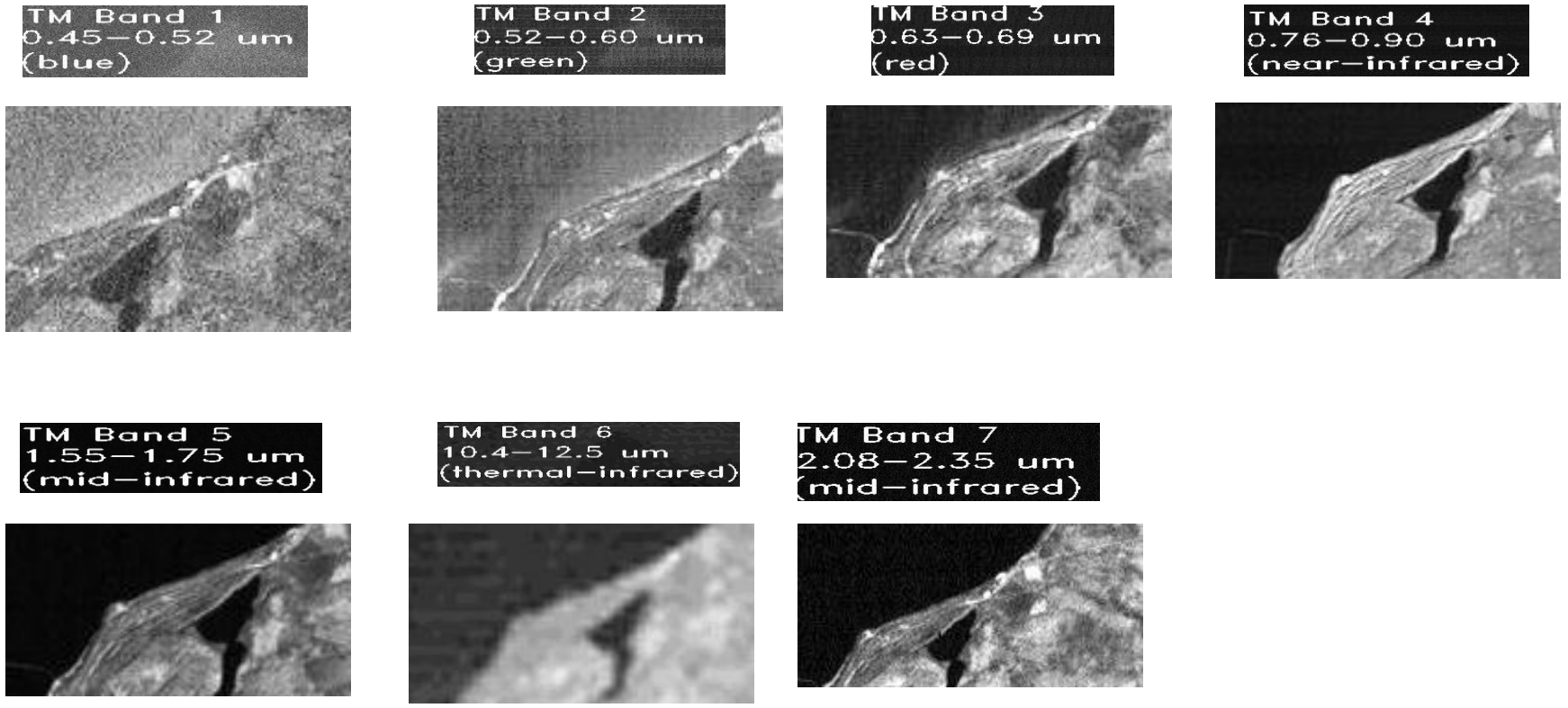


Indus basin from Landsat 30 m

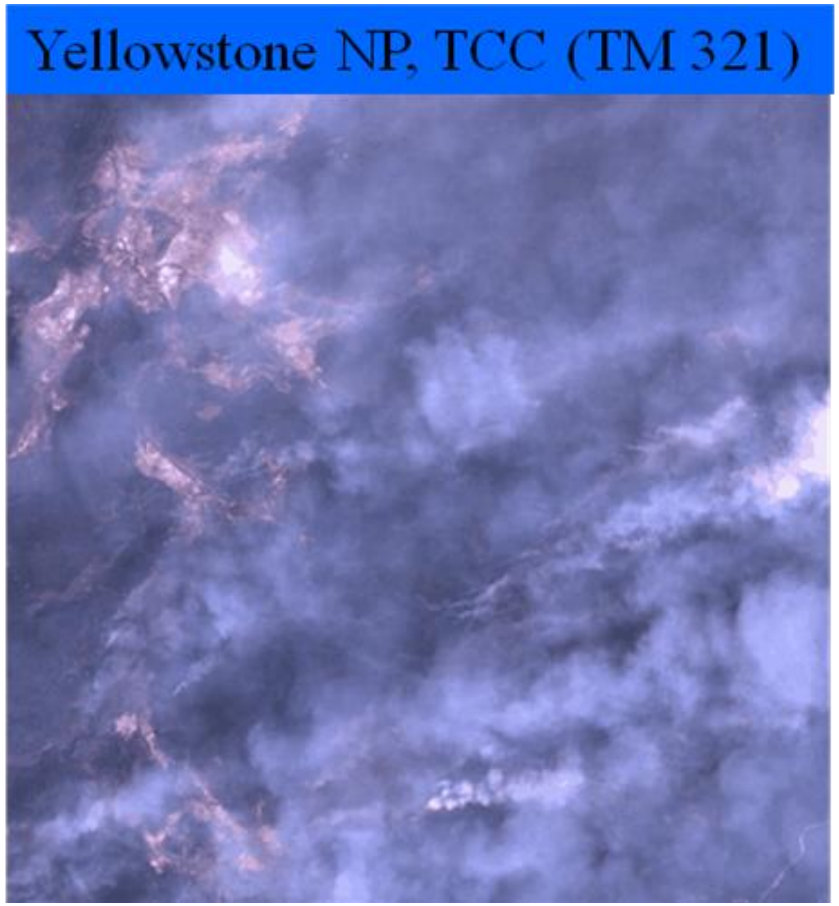


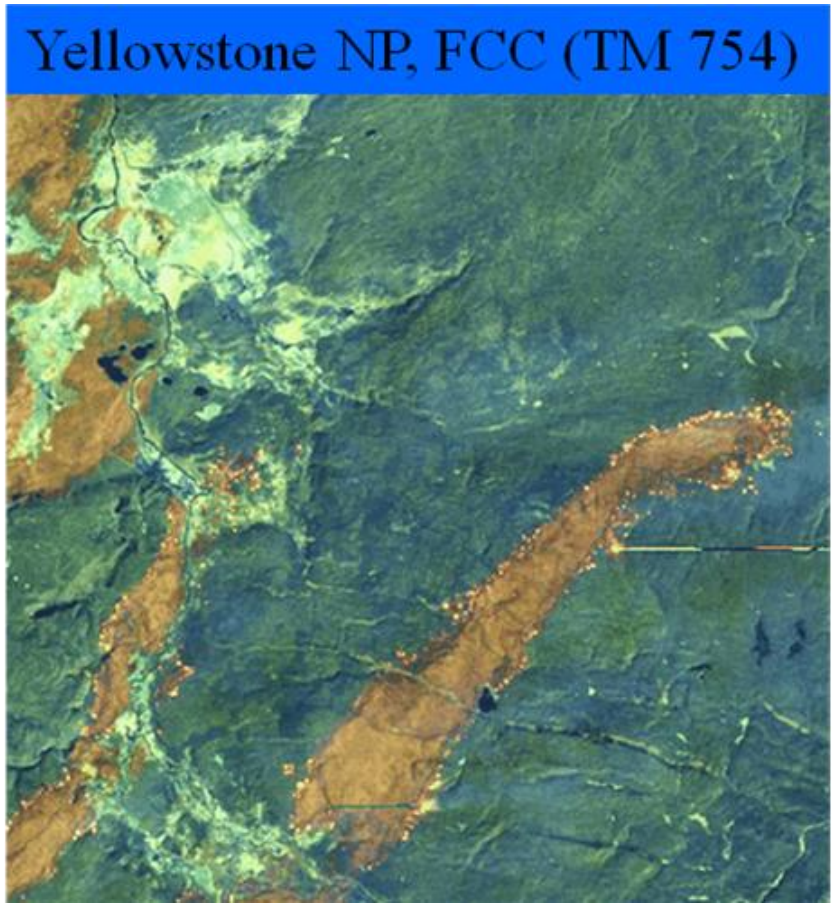
Karkeh sub-basin watershed from IKONOS

Difference in the spectral responses of an area in different bands of Landsat TM image



Forest Fire (Yellowstone NP)





Sensor/system	Platform	Revisit period (day)	No. of bands	Spectral range/band	Spatial resolution (m) MS (optical, TIR)	Pan	Spectral resolution (nm)	Operation
Moderate spatial resolution								
ASTER	Satellite	16	14	Green, red, NIR, 6SWIR, 5TIR	15/30, 90		Broad band	1999-
Landsat-5 TM	Satellite	16	7	Blue, green, red, NIR, 2SWIR, TIR	30, 120		Broad band	1984-2012
Landsat-7 ETM+	Satellite	16	8	Blue, green, red, NIR, 2SWIR, TIR, pan	30, 60	15	Broad band	1999-
Landsat-8 OLI/TIRS	Satellite	16	11	CA, blue, green, red, NIR, 3SWIR, 2TIR, pan	30, 100	15	Broad band	2013-
Sentinel-2A MSI	Satellite	10	13	2Blue, green, yellow, RE, 5NIR, 3SWIR	10/20/60		Broad band	2015-
SPOT-1-5	Satellite	3	4/5	Green, red, NIR, SWIR, pan	10/20	5/10	Broad band	1986-
High spatial resolution								
GeoEye-1	Satellite	2-3	5	Blue, green, red, NIR, pan	1.65	0.41	Broad band	2008-
Gaofen-2	Satellite	5	5	Blue, green, red, NIR, pan	3.2	0.8	Broad band	2014-
IKONOS	Satellite	3	5	Blue, green, red, NIR, pan	4	1	Broad band	1999-2014
Quickbird	Satellite	3-4	5	Blue, green, red, NIR, pan	2.4	0.6	Broad band	2001-
Pléiades	Satellite	1	5	Blue, green, red, NIR, pan	2	0.5	Broad band	2011-
RapidEye	Satellite	1	5	Blue, green, red, RE, NIR	5		Broad band	2008-
WorldView-2/3 (WV2/3)	Satellite	1.0-4.5	9/17	2Blue, green, yellow, red, RE, 2NIR, pan, 8SWIR	2/1.24/3.7	0.5/ 0.31	Broad band	2009-, 2014-
High spectral resolution								
AISA (eagle, hawk, dual)	Airborne		244, 254	400-970 nm, 1000-2500 nm, 400-2500 nm	Variable		1.6-9.0	1993-
AVIRIS	Airborne		224	400-2500 nm	Variable		10	1987-
CASI (CASI-2)	Airborne		288	405-950 nm	Variable		2.2	1990-
CHRIS	Satellite		63	410-1050 nm	18-36		1.25-11.0	2001-
HYDICE	Airborne		206	400-2500 nm	Variable		7.6-14.9	1995-
HyMap	Airborne		126	400-2500 nm	Variable		10-20	1996-
Hyperion, hyperspectral imager	Satellite, EO-1	>15	220	400-2500 nm	30		10	2000-2015

Note: CA—coastal aerosol; RE—red edge; NIR—near infrared; SWIR—shortwave infrared; TIR—thermal infrared; Pan—pancromatic; MS—multispectral.